

Fractional Shapiro steps in a cavity-coupled Josephson ring condensate

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(Received 8 July 2025; accepted 22 October 2025; published 20 November 2025)

The Josephson effect presents a fundamental example of macroscopic quantum coherence as well as a crucial enabler for metrology (e.g., voltage standard), sensing (e.g., superconducting quantum interference device), and quantum information processing (Josephson qubits). Recently, there has been a major renewal of interest in the effect, following its observation in Bose, Fermi, and dipolar atomic condensates, in exciton-polariton condensates, and in momentum space. We present theoretically a nondestructive, *in situ*, and real time protocol for observing the ac and dc Josephson effects including integer (recently observed in cold atoms) and fractional (hitherto unobserved in cold atoms) Shapiro steps, using a ring condensate coupled to an optical cavity. Our analysis presents a single-shot metrology standard that does not require measurement of atomic number and that challenges the conventional wisdom that quantum computations cannot be observed without being destroyed. Our results have implications for the fields of atomtronics, sensing, metrology, and quantum information processing.

DOI: [10.1103/PhysRevResearch.7.043188](https://doi.org/10.1103/PhysRevResearch.7.043188)

I. INTRODUCTION

The Josephson effect [1], first proposed to describe the emergence of particle current between weakly coupled superconductors [2], has been a topic of immense interest in the condensed matter and quantum optics communities both as an exemplar of macroscopic quantum coherence and as an enabler of applications in metrology and quantum information processing [3–9]. Subsequently, the effect was observed in an array of reservoirs connected via submicron apertures in superfluid ³He [10]. There has been a renewed interest in the topic from the ultracold atom community following the realization of the bosonic Josephson junction [11–16], which differs from its superconductor and superfluid counterparts due to its high experimental tunability and ease of theoretical characterization. Recently, Josephson effects have also been observed in supersolids [17–19] and in momentum space [20,21] and studied theoretically in various configurations using the Gross-Pitaevskii equation and Bogoliubov analysis [14,22–25]. A key finding in this context is the emergence of Shapiro steps—a hallmark of phase synchronization between two reservoirs and an externally driven Josephson junction [26]. These have been observed in both Bose and Fermi systems [27–30], representing the accession of the voltage standard to the platform of ultracold quantum gases. However,

the observation of fractional Shapiro steps, already performed in superconducting systems [31–40], has not yet been made in ultracold quantum gas systems.

Specifically, inspired by the superconducting quantum interference devices (SQUIDs) [41], atomic quantum interference devices (AQUIDs) [15,42–44] have been developed using a Bose-Einstein condensate (BEC) confined to an annular trap, which naturally supports the atomic supercurrent while preserving the coherence of the condensate phase [45–52]. An AQUID typically consists of one or more Josephson junctions, in the form of optical tunneling barriers, allowing observation of phase slips [50,53], the dc-ac Josephson effect [13,15], quantum interference of currents [44,53,54], parity-protected qubits [55], and stabilization of persistent currents [56]. Recently, the realization of the Josephson ring junction in an exciton-polariton condensate has opened the door to applications at room temperature [57].

These demanding applications of cold-atom Josephson systems in general and AQUIDs in particular require precise information on the motional dynamics of the condensate. To date, all realized (see above) and proposed [54] detection of current states in cold-atomic Josephson junctions has been totally destructive, requires time-of-flight expansion of the atomic gas, and cannot be carried out in real time. These restrictions require repeated sample preparation, expose the system to variations in initial conditions, slow down metrological measurements, and forbid checks during quantum computations.

To remove all these restraints, we theoretically propose a nondestructive, *in situ*, and real-time method of measuring Josephson oscillations. Our protocol is based on our previous work on the detection of persistent currents and solitons [58–60], later extended to the detection of super-

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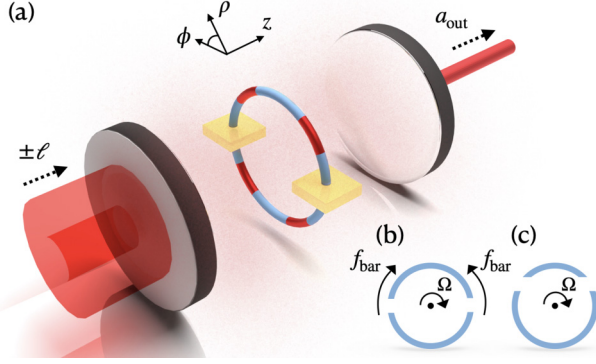


FIG. 1. (a) Schematic diagram of proposed experimental setup. A ring-trapped BEC with two Josephson junctions (JJs) is placed inside an optical cavity. The Josephson junction dynamics is probed by two LG beams carrying orbital angular momenta $\pm\ell\hbar$ that form an annular lattice (red shaded region). a_{out} denotes the optical field transmitted by the cavity. (b) Initial setup with the JJs that are placed symmetrically across the ring, resulting in two half rings. The JJs are rotated at angular frequency Ω , and simultaneously moved toward each other with velocity f_{bar} up to a time interval t_{bar} to generate a chemical potential difference between the two regions. (c) Final positions of the JJs beyond t_{bar} .

fluid drag [61], and exploits the sensitive techniques of cavity optomechanics [62,63]. In this work, we consider the much more complicated case of a ring condensate with Josephson junctions coupled to an optical cavity driven by orbital angular momentum (OAM)-carrying beams. In spite of the highly multimodal nature of the dynamics, we show how our technique can yield clean signatures, in the cavity transmission, of the dc and ac Josephson effects as well as the Shapiro steps. Although for reasons of simplicity we restrict our present demonstration to AQUIDs, our technique of detecting Josephson oscillation can, in principle, be generalized to other condensate geometries that preserve quantized circulation, such as a Josephson junction necklace (ring condensate with more than two junctions) or a multicomponent ring condensate.

Our work unfolds as follows. In Sec. II, we present the theoretical model, simulation setup, mean-field model approach, and optical barrier. Section III discusses the detection of the Josephson oscillation frequency, followed by the dc-to-ac Josephson transition in Sec. IV. In Sec. V, we explore the emergence of integer and fractional Shapiro steps. Finally, we conclude our work in Sec. VI. Appendix A provides the details of AQUID readout using the cavity output spectra, Appendix B compares the resistively-capacitively shunted junction (RCSJ) model with the Gross-Pitaevskii equation (GPE) simulation, and Appendix C analyzes the fractional Shapiro steps across various conditions.

II. THEORETICAL MODEL

The method requires an annularly magnetically trapped condensate with optically enforced Josephson junctions to be kept at the center of a high-finesse optical cavity, as shown in Fig. 1. Driving the cavity with a superposition of two Laguerre-Gaussian beams of OAM $\pm\ell\hbar$ [64] leads to a weak

angular optical lattice [65]. When a rotating matter wave with winding number L_p interacts with this lattice, it undergoes Bragg diffraction into rotational states $L_p \pm 2n\ell$, where n is an integer. These currents serve as the mechanical modes optomechanically interacting with the intracavity field [58,63]. The cavity field is in turn modulated at the condensate's side-mode as well as Josephson oscillation frequencies, which can be obtained by a homodyne detection of the cavity-output field [62].

We numerically model the proposed experimental setup by considering a one-dimensional ring BEC of ^{23}Na atoms with two Josephson junctions placed inside an optical cavity. To probe the mechanical oscillation of BEC, an optical lattice potential is formed by the superposition of two driving LG beams with charge $\pm\ell$. In the frame rotating with the laser drive frequency, the azimuthal motion of the BEC is described by the many-body Hamiltonian [58,59]

$$H = \int_0^{2\pi} d\phi \Psi^\dagger(\phi) \left[\mathcal{H}^{(1)} + \frac{g}{2} \Psi^\dagger(\phi) \Psi(\phi) \right] \Psi(\phi) + \mathcal{H}_c, \quad (1)$$

where $\Psi(\phi)$ is the bosonic atomic field operator obeying $[\Psi(\phi), \Psi^\dagger(\phi')] = \delta(\phi - \phi')$, g denotes the two-body atomic interaction strength, and $\mathcal{H}^{(1)}$ represents the single-atom Hamiltonian

$$\mathcal{H}^{(1)} = -\frac{d^2}{d\phi^2} + V_{\text{opt}} + V_{\text{bar}}, \quad (2)$$

which includes the rotational kinetic energy and potential energy contribution from the optical lattice and the optical barriers. We have presented the detailed derivation of these terms in Secs. II A and II B, respectively. The last term in Eq. (1) includes the energy contributions from cavity field and laser drive, expressed as

$$\mathcal{H}_c = -\hbar\Delta_o a^\dagger a + i\hbar\eta(a^\dagger - a), \quad (3)$$

respectively, where a is the photonic operator that obeys $[a, a^\dagger] = 1$, Δ_o is the detuning of driving field frequency from the cavity resonance frequency ω_0 , and $\eta = \sqrt{P_{\text{in}}}\gamma_0/\hbar\omega_0$ is the pump rate expressed in terms of input optical power P_{in} and cavity decay rate γ_0 .

A. Optical lattice potential

By treating condensate atoms inside the cavity as a two-level system and in the limit of large atom-photon detuning, the dispersive light-matter interaction is governed by [58,59]

$$V_{\text{opt}} \approx \frac{\hbar\mathcal{D}_{eg}^2\mathcal{E}_0^2|u_{+\ell 0}(\mathbf{r}) + u_{-\ell 0}(\mathbf{r})|^2}{\Delta_a} a^\dagger a, \quad (4)$$

where $a^\dagger a$ is the photon number of the cavity mode, $\mathcal{D}_{eg}$ is the transition dipole moment operator between the atomic ground and excited states, Δ_a is the detuning of the driving laser from the atomic transition frequency, $\mathcal{E}_0$ denotes the single-photon electric field amplitude inside the cavity, $u_{\pm\ell 0}(\mathbf{r})$ are the mode functions of the LG beams of OAM $\pm\ell\hbar$, and $|u_{+\ell 0}(\mathbf{r}) + u_{-\ell 0}(\mathbf{r})|^2$ gives the cavity mode intensity distribution at atomic position r . Near $z = 0$ plane, the intensity distribution of the superposition simplifies to $\approx \cos^2(\ell\phi)$. By grouping constant terms into an effective coupling strength,

we get

$$V_{\text{opt}} = \hbar U_0 \cos^2(\ell\phi) a^\dagger a. \quad (5)$$

B. Optical barrier potential

The optical barrier potential, which can be created by “painting” a repulsive focused beam onto the condensate, is modeled as [56]

$$V_{\text{bar}} = V_0 \left[e^{-2(\frac{\phi-\phi_{b1}}{\sigma})^2} + e^{-2(\frac{\phi-\phi_{b2}}{\sigma})^2} \right], \quad (6)$$

where ϕ_{b1} and ϕ_{b2} are the positions of two barriers that act as weak links. In simulation, their heights V_0 and widths σ are considered of the order of the chemical potential without any barriers μ_0 and condensate healing length ζ , respectively.

C. Equations of motion

Following the system Hamiltonian as described in Eq. (1), the coupled dynamics of the condensate wave function $\psi(\phi, t)$ and cavity field amplitude α are described by the following set of nondimensionalized equations [58–60,62,63]:

$$\begin{aligned} (i - \Gamma) \frac{d\psi}{d\tau} = & \left[\left(-i \frac{d}{d\phi} - \Omega' \right)^2 + \frac{U_0}{\omega_\beta} |\alpha(\tau)|^2 \cos^2(\ell\phi) \right. \\ & \left. + V_{\text{bar}}(\phi, \tau) - \mu + \mathcal{G} |\psi(\phi, \tau)|^2 \right] \psi \\ & + \xi(\phi, \tau), \end{aligned} \quad (7)$$

$$\begin{aligned} i \frac{d\alpha}{d\tau} = & \left\{ - \left[\Delta_c - U_0 \langle \cos^2(\ell\phi) \rangle_\tau + i \frac{\gamma_0}{2} \right] \alpha + i\eta \right\} \omega_\beta^{-1} \\ & + i\sqrt{\gamma_0 \omega_\beta^{-1}} \alpha_{\text{in}}(\tau). \end{aligned} \quad (8)$$

We employ the dimensionless formulation by scaling energy as $\hbar\omega_\beta = \hbar^2/2mR_0^2$ and time as $\tau = \omega_\beta t$. Here, m is the mass of the atom and R_0 is the radius of the ring trap, also used as the unit of length ($\phi = x/R_0$). The condensate wave function is normalized to the total number of atoms via $\int |\psi(\phi, t)|^2 d\phi = N$. The terms of Eqs. (7) and (8) are described in detail below. The first term on the right-hand side of Eq. (7) represents the rotational kinetic energy of the atoms in the frame corotating with barriers, with rotation frequency $\Omega = \Omega' \times (2\omega_\beta)$. The second term describes the optical lattice potential that couples the condensate to the cavity field with strength $U_0 = g_a^2/\Delta_a$, where g_a is the single-atom-photon coupling strength and Δ_a is the detuning of the optical field from the atomic transitions. The third term represents the optical barrier potential, and the application of the two optical barriers leads to the formation of two half-ring condensates separated by two JJs. A similar experimental setup has previously been demonstrated to observe the Josephson effect in a ring BEC [15,56]. The total number of atoms is distributed between the two half-rings as $N_j = \int_{S_j} |\psi|^2 d\phi$, where the integration is performed over the respective regions. The first region corresponds to $S_1 : (\phi_{b1} \leq \phi < \phi_{b2})$ and the second region corresponds to $S_2 : (\phi < \phi_{b1} \text{ or } \phi > \phi_{b2})$. The fourth term on the right-hand side of Eq. (7) denotes the chemical potential of the condensate, which has been updated in each time step to conserve the norm of the wave function [66] in the presence of the dissipation Γ and thermal fluctuation ξ .

The fifth term accounts for the two-body atomic interaction with strength $\mathcal{G} = g/\hbar\omega_\beta$, where $g = 2\hbar\omega_\rho a_s/R_0$, with the s -wave scattering length a_s and the harmonic trap frequency along the radial direction ω_ρ .

The first term on the right-hand side of Eq. (8) represents the detuning of the driving laser field from the cavity resonance frequency ($\Delta_c = \omega_L - \omega_c$), which governs the intracavity field buildup and thus controls the dynamical response of the system. The second term contains the expectation value of the optical lattice potential, which describes the coupling between the cavity field and the condensate. In the third term, γ_0 denotes the cavity linewidth and $\eta = \sqrt{P_{\text{in}} \gamma_0 / \hbar \omega_c}$ is the pump rate, where P_{in} is the input optical power.

The final terms of both Eqs. (7) and (8) account for the thermal fluctuations in the condensate and the optical shot noise in the cavity field, respectively. These stochastic contributions are described by delta-correlated white noise [58,67] i.e.,

$$\langle \xi(\phi, \tau) \xi^*(\phi', \tau') \rangle = \frac{2\Gamma k_B T}{\hbar\omega_\beta} \delta(\phi - \phi') \delta(\tau - \tau'), \quad (9)$$

$$\langle \alpha_{\text{in}}(\tau) \alpha_{\text{in}}^*(\tau') \rangle = \omega_\beta \delta(\tau - \tau'), \quad (10)$$

where k_B is the Boltzmann constant and T is the temperature of the BEC. In the simulation, these Langevin noise terms are modeled as

$$\xi(\phi, \tau) = \sqrt{2\Gamma k_B T / (\hbar\omega_\beta d\phi d\tau)} \mathcal{N}(0, 1, N_\phi), \quad (11)$$

$$\alpha_{\text{in}}(\tau) = \sqrt{\omega_\beta / d\tau} \mathcal{N}(0, 1, 1). \quad (12)$$

The term $\mathcal{N}(0, 1, N_\phi)$ in Eq. (11) consists of N_ϕ normally distributed random numbers with zero mean and unit variance, where N_ϕ is the number of grid points in the spatial coordinate ϕ . Here, N_ϕ random numbers are used to match the size of the condensate wave function array in the simulation. However, in the case of laser shot noise, the array of $\alpha(\tau)$ requires only one random number to be added in each time step. For this reason, we use $\mathcal{N}(0, 1, 1)$ in Eq. (12). The noise prefactor in Eq. (11) includes the phenomenological damping parameter Γ according to the fluctuation-dissipation theorem and $1/(d\phi d\tau)$ arises due to the discretisation of the Dirac-delta function in the noise correlation.

In all our simulations, the first step is to obtain the ground state of the condensate in the presence of optical barriers, which is done by evolving a uniform density state with a random phase profile in imaginary time. The achieved ground state is then evolved in real time up to time t_{bar} , when the barriers are moving to generate a dc or an ac atomic current. Barrier motion is achieved by changing the positions of barriers, which are included in V_{bar} in each time frame. Once the barrier motion stops, the cavity field is turned on and the coupled BEC-cavity equations are solved up to time $t_{\text{sim}} = 3$ s. This ensures that the cavity field captures the dynamics of the Josephson oscillation under the condition of static barriers, without the additional contribution from the transient driving effect associated with the barrier motion.

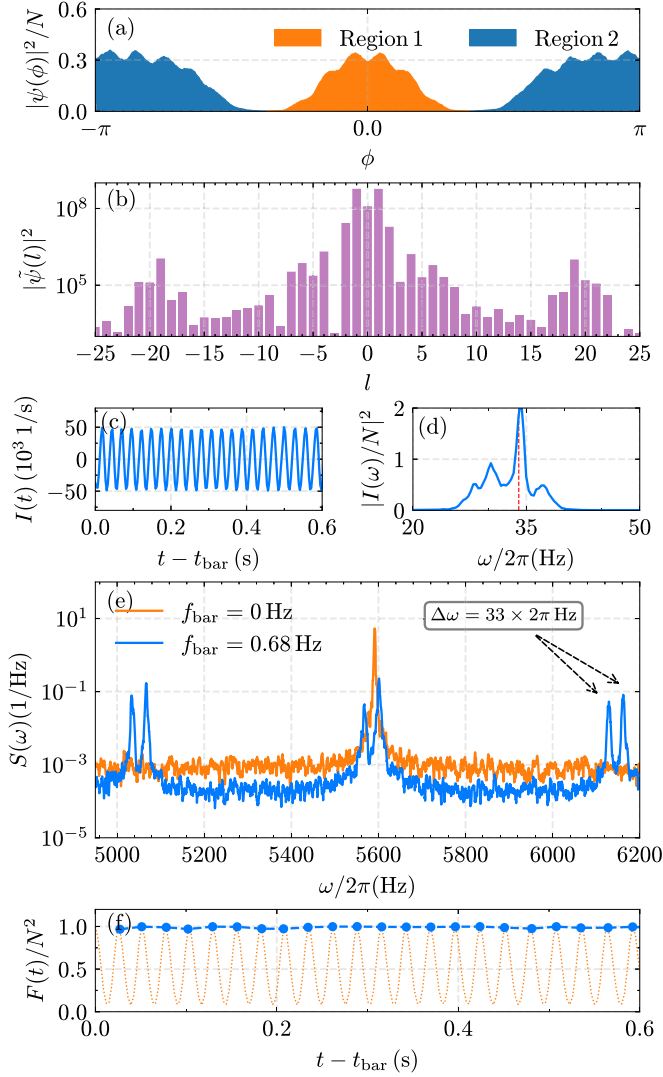


FIG. 2. Numerical results obtained after solving Eqs. (7) and (8): (a) Density profile of the condensate wave function obtained at the end of the simulation. (b) The OAM distribution of the corresponding condensate wave function. (c) Josephson tunneling current across the barrier. (d) Power spectrum of Josephson tunneling current. The dominant peak corresponds to the Josephson oscillation frequency $\omega_J \sim 33 \times 2\pi$ Hz as shown with a red dashed line. (e) Power spectrum of the phase quadrature of the cavity output field. (f) Temporal evolution of fidelity of condensate wave function. The blue dots connect the highest points in the fidelity curve (orange line) that oscillates at the Josephson tunneling frequency ω_J . The parameter values used are $m = 23$ amu, $N = 8100$, $R_0 = 4$ μm , $f_{\text{bar}} = 0.68$ Hz, $V_0 = 7.2 \mu_0 = 23$ nK, $\sigma = 1.2 \zeta = 2 \mu\text{m}$, $\zeta = 1/\sqrt{\mu_0/\hbar\omega_\beta}$, $t_{\text{bar}} = 0.092$ s, $\Gamma = 0.0001$, $T = 10$ nK, $P_{\text{in}} = 10$ fW, $U_0 = 2\pi \times 212$ Hz, $\tilde{\Delta} = \Delta_0 - U_0 N/2 = -2\pi \times 173$ Hz, $\Delta_a = 2\pi \times 4.7$ GHz, $\omega_c = 2\pi \times 10^{15}$ Hz, $\omega_\rho = \omega_z = 2\pi \times 42$ Hz, $\gamma_0 = 2\pi \times 2$ MHz, and $\Omega' = 0$ [42,50].

III. DETECTION OF JOSEPHSON OSCILLATION FREQUENCY

To demonstrate the detection method, we show the numerical results obtained by solving Eqs. (7) and (8) for a particular barrier velocity $f_{\text{bar}} = 0.68$ Hz in Fig. 2. This bar-

rier movement results in compression of atoms in one region, as shown in Fig. 2(a), and this compression results in a chemical potential difference $\Delta\mu$ between the two regions. Now, the phase difference across the junction ($\Phi = \Phi_2 - \Phi_1$) evolves as $\frac{d\Phi}{dt} = -\frac{\Delta\mu}{\hbar}$ [13,15]. This triggers the Josephson oscillation in the population imbalance ($n = N_2 - N_1$) at the frequency $\omega_J = \Delta\mu/\hbar$. To characterize the rotational state of the condensate, we have calculated the OAM distribution ($|\tilde{\psi}(l)|^2$) and shown in Fig. 2(b). Here, $\tilde{\psi}(l)$ is the Fourier transformation of the condensate wave function $\psi(\phi)$. The dominance of $l = 0, \pm 1$ modes indicates the quantum superposition of persistent current states induced by the moving barriers [54,55,68]. Denoting the winding number of these states as L_p , they Bragg scatter to $L_p \pm 2\ell$ states in the presence of the weak optical lattice potential [58]. The Josephson tunneling current flowing across the JJs is calculated by numerically differentiating the population imbalance: $I = \frac{dn}{dt}$ [13,15] and shown in Fig. 2(c). Performing a Fourier transformation of this tunneling current allows us to calculate the oscillation frequency [13], which is $\omega_J = 33 \times 2\pi$ Hz for this case [see Fig. 2(d)]. Remarkably, as we propose, this Josephson oscillation frequency can be directly obtained from the power spectrum of the phase quadrature of the cavity output field ($S(\omega) = |\text{Im}[\alpha_{\text{out}}(\omega)]|^2$), where $\alpha_{\text{out}}(t)$ is calculated numerically by using the input-output relation $\alpha_{\text{out}} = -\alpha_{\text{in}} + \sqrt{\gamma_0}\alpha$ [62]. We emphasize that our method does not require any imaging technique to measure the number of atoms in each region, or destruction of the condensate, as needed in previously available methods for studying Josephson dynamics in ultracold quantum gases [13,15,28,44,69].

We present the power spectrum of the phase quadrature of the cavity output field as a function of the system response frequency for the above-mentioned case in Fig. 2(e). We see the condensate sideband peaks split for junction velocity $f_{\text{bar}} = 0.68$ Hz, which contrasts with the case when the junctions do not move ($f_{\text{bar}} = 0$ Hz). The reason for this splitting is due to the generation of a chemical potential difference that triggers the Josephson oscillation in the population imbalance. This population oscillation modulates the light field, creating additional sidebands and splitting the peaks in the spectrum by a magnitude $\Delta\omega$ equal to the Josephson frequency ω_J . Importantly, both the spectral peaks show the same magnitude of splitting, highlighting that the splitting originates from a single coherent Josephson tunneling. So, by observing these splits in the cavity output spectrum, the Josephson oscillation frequency can be detected without destroying the condensate or measuring the atom number. To ascertain the efficacy of our proposed scheme, we compute the fidelity of condensate wave function over time, defined as $F(t) = \int_{-\pi}^{\pi} [\psi^*(\phi, t)\psi(\phi, 0)]^2 d\phi$, and present it in Fig. 2(f). We find that the fidelity remains close to unity, evidencing the nondestructive nature of the measurement technique proposed in this work.

IV. DETECTION OF DC TO AC JOSEPHSON TRANSITION

We demonstrate the detection of the transition from the dc to ac Josephson regime by varying the junction velocity (f_{bar}), as shown in Fig. 3. The time evolution of condensate density and particle current during the barrier movement is

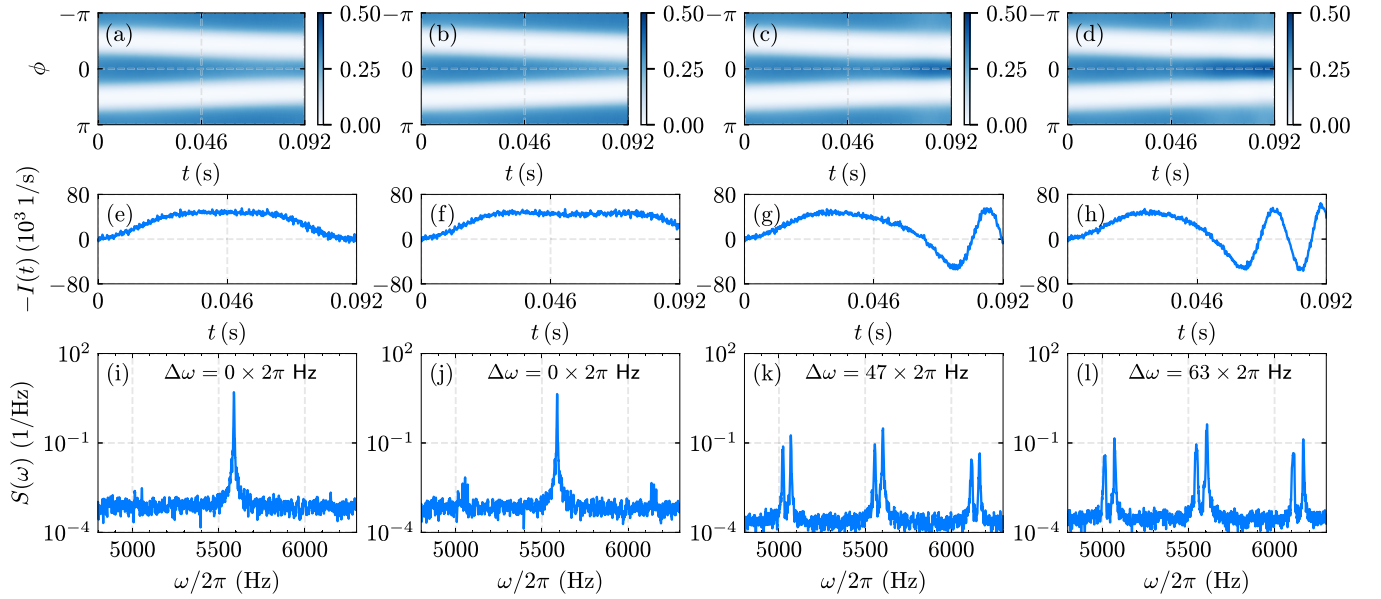


FIG. 3. dc-ac Josephson transition: (a)–(d) Time evolution of condensate density during the barrier movement. (e)–(h) Time evolution of particle current during the junction movement. (i)–(l) Power spectra of phase quadrature of the cavity output field vs the system’s response frequency. Here, junction velocities are $f_{\text{bar}} = 0.6, 0.65, 0.7$, and 0.75 Hz, respectively. The other set of parameters is the same as used in Fig. 2.

presented in Figs. 3(a)–3(d) and 3(e)–3(h), respectively. For barrier velocity f_{bar} below a critical value f_c , the condensate densities in the two regions remain the same due to the coherent tunneling of atoms across the JJs maintaining zero chemical potential difference [Figs. 3(a) and 3(b)]. Beyond a critical barrier velocity, which corresponds to the critical current of the JJs, is reached, we see the compression (expansion) of atoms in region 1 (region 2) corresponding to $\Delta\mu \neq 0$ [Figs. 3(c) and 3(d)]. This manifests as the Josephson oscillation of population imbalance as seen in Figs. 3(g) and 3(h).

As demonstrated in Fig. 2, the onset of nonzero chemical potential difference gives rise to a measurable splitting in the peaks in the cavity output spectrum, with the amount of splitting revealing the Josephson oscillation frequency. These spectral features are well above the noise floor, allowing us to measure them precisely. The magnitude of the splitting $\Delta\omega$ further increases with an increase in the junction velocity f_{bar} as seen in Figs. 3(k), 3(l), and 4(a). The value of the junction velocity for which $\Delta\omega$ becomes nonzero and further increases is denoted as the critical barrier velocity f_c , which is 0.68 Hz in our case. Summarizing these observations, we distinguish between the dc and ac Josephson regimes by the presence or absence of peak splitting ($\Delta\omega$) in the cavity output spectrum, respectively. In the inset of Fig. 4(a), we show the variation of $\Delta\omega$ with the rotation rate Ω' of AQUID, which oscillates with a period corresponding to one flux quantum $\hbar/mR_0^2$ (the relevant cavity output spectra are shown in the Appendix). This particular feature demonstrates that our system enables nondestructive detection of quantum interference of clockwise and anticlockwise currents [44,54], which can be effectively utilized as a quantum rotation sensor analogous to a SQUID magnetometer [41] as well as a qubit in quantum information processing [68].

V. OBSERVATION OF SHAPIRO STEPS

In the bosonic JJ, Shapiro steps appear as plateaus in chemical potential difference ($\Delta\mu$) at integer multiples of external driving frequency [27,28]. The external drive is applied through a combination of linear translation (dc) and periodic modulation (ac) of junction position, effectively mimicking the external microwave radiation applied to superconducting JJs [26]. To observe the Shapiro steps through our setup (AQUID geometry), we periodically drive JJs’ positions along with the linear translations up to t_{bar} . In the frame rotating at Ω , we model the JJs’ positions as $\phi_{b1,2} = \pm(2\pi f_{\text{bar}})t \pm \phi_m \sin(2\pi f_m t)$, where f_{bar} is the junction velocity corresponding to linear translation, f_m is the modulation frequency, and ϕ_m is the modulation amplitude. Then, we solve Eqs. (7) and (8) to obtain the cavity output spectra and subsequently the peak splitting $\Delta\omega$ that acts as the DC voltage in the SQUID. We plot the variation of $\Delta\omega$ for various barrier velocities f_{bar} in Fig. 4(b) and see the emergence of Shapiro steps at integer multiples of modulation frequency f_m for $\Omega' = 0$. The width of the Shapiro steps can be further increased by increasing the modulation amplitude ϕ_m [27,28]. It is worthwhile to mention at this point that by rotating the AQUID device, which can be achieved by rotating the barriers, fractional Shapiro steps appear between the integral steps. It is analogous to the observation of half-integer Shapiro steps in dc SQUID exposed to an external magnetic field [31,32]. However, for half-integer flux quantum, when the system is in a balanced superposition of adjacent momentum states [54,68], the phase difference at two junction evolves differently with time due to rotation and one junction exhibits additional 2π rotation than that of the other, resulting in flipflop between adjacent winding states [31]. As a result of this, the total voltage becomes $(n + \frac{1}{2}) f_m$, which gets manifested in the form

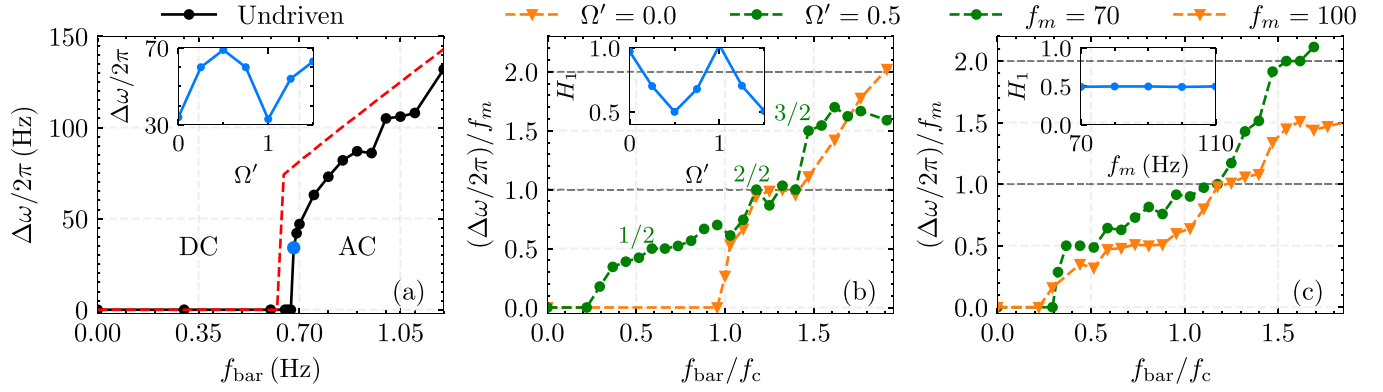


FIG. 4. Variation of the amount of peak splitting ($\Delta\omega$) in the cavity output spectra as a function of the barrier velocity (f_{bar}) for (a) $f_m = 0$ Hz, $\phi_m = 0$, $\Omega' = 0$, (b) $f_m = 90$ Hz, $\phi_m = 0.01$, and (c) $\Omega' = 0.5$, $\phi_m = 0.01$. The red dashed line in panel (a) represents the results obtained from the RCSJ model (see details in the Appendix). The inset in panel (a) shows the oscillation in $\Delta\omega$ with the rotation rate of AQUID for $f_{\text{bar}} = 0.68$ Hz. The inset in panel (b) shows the variation of the first step height H_1 with AQUID rotation rate Ω' . The inset in panel (c) shows the first step height H_1 for different modulation frequencies f_m . The variation of $\Delta\omega$ as a function of f_{bar} for these cases is shown in the Appendix. The other set of parameters is the same as used in Fig. 2.

of plateau at exactly half-integer multiples of the modulation frequency.

VI. CONCLUSION

We have proposed and numerically simulated a cavity optomechanical technique for the single-shot nondestructive measurement of the Josephson oscillation frequency in an atomtronic quantum interference device. The cavity output spectra reveal the Josephson oscillation frequency precisely as a resolvable splitting in the side-mode peaks, without requiring the challenging measurement of atom number. The onset of the dc to ac Josephson transition and emergence of the integer (recently detected in cold atoms) and fractional (yet to be detected in cold atoms) Shapiro steps are directly observed through these spectral features, offering a clear and experimentally accessible signature of the Josephson tunneling dynamics. In the present work, we have mainly focused on detecting the half-integer Shapiro steps. However, other fractions can be readily accessed with asymmetric barriers or modulation with higher harmonics. Our approach highlights the potential of cavity optomechanics as a sensitive, nondestructive probe for atomtronic quantum interference devices and facilitates progress in metrology, quantum computing, and information processing with ultracold atoms.

ACKNOWLEDGMENTS

We gratefully acknowledge the supercomputing facilities Param Ishan and Param Kamrupa, where all the simulation runs were performed. M.B. thanks the Air Force Office of Scientific Research (FA9550-23-1-0259) for support. R.K. was supported by JSPS KAKENHI (Grant No. 25K07190) and JST ERATO (Grant No. JPMJER2302).

DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

APPENDIXES

Here, we provide the demonstration of the rotation-induced effect on the cavity output spectra, details of the RCSJ model, and its comparison with GPE simulation results.

In Appendix A, we demonstrate the readout of an AQUID. In Appendix B, we analyze a dc SQUID circuit by calculating the output voltage across it using the RCSJ model and compare it to the results obtained via simulating the Gross-Pitaevskii equation for the AQUID. Finally, in Appendix C, we demonstrate the emergence of the Shapiro steps in the voltage-current curve of AQUID using GPE simulation and compare the results with the RCSJ model predictions, and both the results match quite well.

APPENDIX A: CAVITY OPTOMECHANICAL READOUT OF AQUID

We consider an AQUID that consists of two Josephson junctions (JJ_1 and JJ_2). A schematic diagram of this setup is illustrated in Fig. 5. The JJs are moved toward each other to generate a bias current I , which results in the Josephson tunneling of atoms. To emulate the effect of an external magnetic

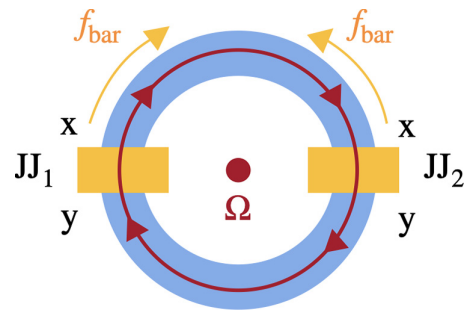


FIG. 5. Schematic diagram of an AQUID that contains two Josephson junctions (JJ_1 and JJ_2) with rotation frequency Ω . The JJs are moved toward each other with velocity f_{bar} to generate the chemical potential difference (bias voltage) and a bias current $I = f_{\text{bar}}(I_c/f_c)$.

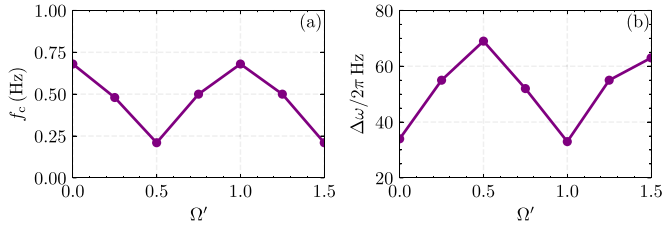


FIG. 6. (a) Variation of critical barrier velocity f_c with AQUID rotation rate Ω' and (b) variation of magnitude of peak splitting $\Delta\omega$ for $f_{\text{bar}} = 0.68$ Hz. Here, $f_m = 0$ Hz, $\phi_m = 0$, and the other parameters used are the same as Fig. 4(a).

field in the case of SQUID, we rotate the JJs around the ring with frequency Ω . In this case, due to the single-valuedness of the wave function, the phase satisfies the relation as [44]

$$\oint \Delta\theta \cdot dl = \theta_{2x} - \theta_{1x} + \theta_{1y} - \theta_{2y} = 2\pi \frac{\Omega}{\Omega_0}. \quad (\text{A1})$$

Considering phase difference across $\text{JJ}_{1,2}$ as $\delta_{1,2}$, using the normalized rotation rate $\Omega' = \Omega/\Omega_0$, where $\Omega_0 = \hbar/mR^2$, one can write

$$\delta_1 = \delta_0 + \pi\Omega', \quad \delta_2 = \delta_0 - \pi\Omega'. \quad (\text{A2})$$

Using the above expressions [Eq. (A2)], the total current in the AQUID can be represented as

$$\begin{aligned} I &= I_1 + I_2 = I_c [\sin(\delta_1) + \sin(\delta_2)] \\ &= I_c [\sin(\delta_0 + \pi\Omega') + \sin(\delta_0 - \pi\Omega')] \\ &= 2I_c \cos(\pi\Omega') \sin(\delta_0). \end{aligned} \quad (\text{A3})$$

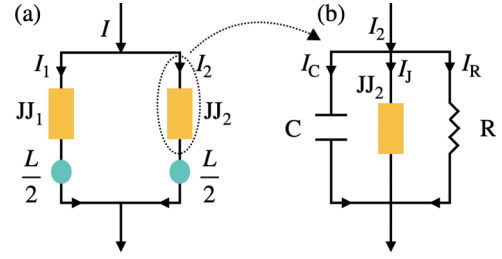


FIG. 8. (a) dc SQUID circuit with two Josephson junctions $\text{JJ}_{1,2}$ and inductance L . (b) RCSJ circuit for a single junction with resistance R and capacitance C .

From Eq. (A3), we find that the critical current of the AQUID $|2I_c \cos(\pi\Omega')|$ varies periodically with the barrier rotation rate Ω with a period Ω_0 . This modulation is a clear signature of interference of currents in the two arms of AQUID [44].

To detect this interference, we plot the critical barrier velocity f_c for different rotation rates Ω' in Fig. 6(a). Here, it is to be noted that the critical barrier velocity is determined by plotting the magnitude of peak splitting in the cavity output spectrum for increasing barrier velocity f_{bar} and the f_{bar} , which indicates the onset of nonzero splitting (or the transition from dc to ac Josephson regime), is denoted as the critical barrier velocity f_c . We can see a clear modulation of the critical barrier velocity (i.e., critical current) with the AQUID rotation rate, with a periodicity of one flux quantum of the ring $\Omega_0 = \hbar/mR^2$.

An easier way to detect the interference of currents is to track the amount of peak split in the cavity output spectra

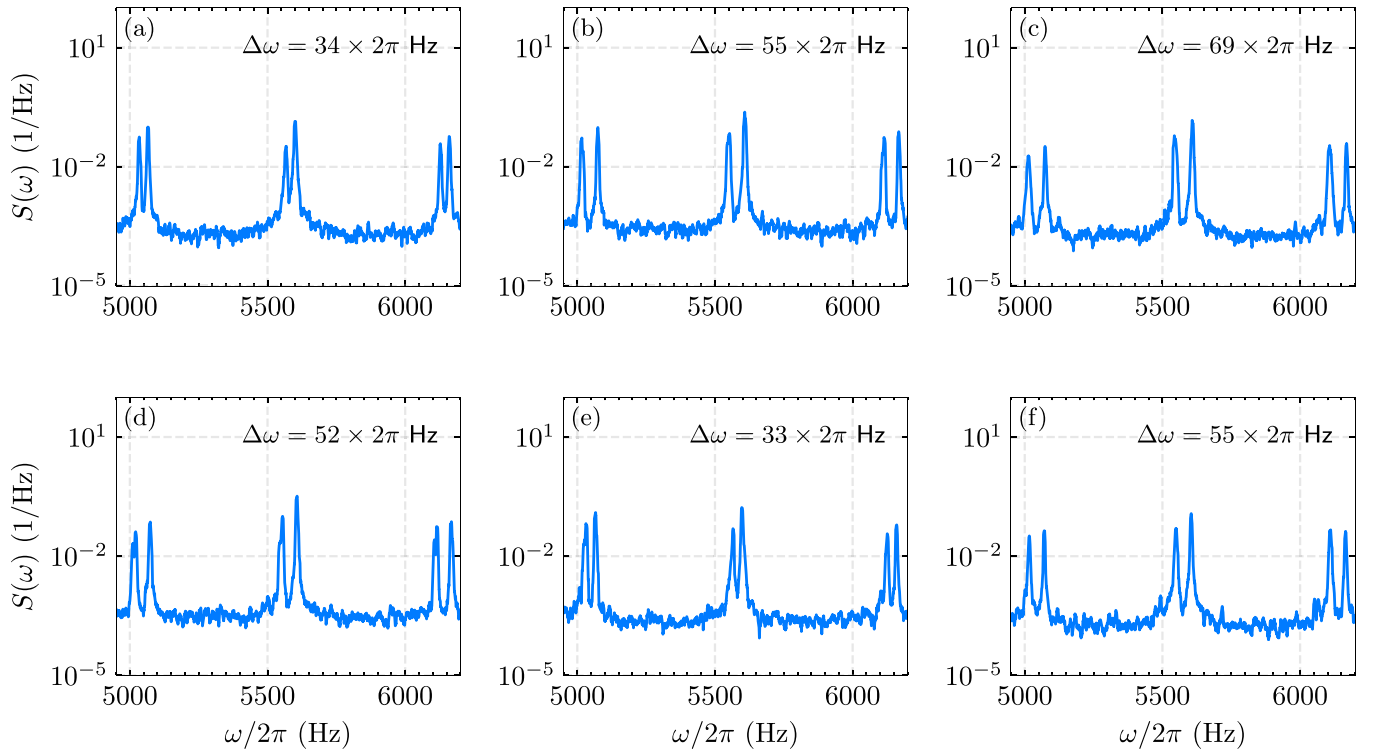


FIG. 7. Power spectrum of the phase quadrature of the cavity output field for junction rotation frequency (a) $\Omega' = 0.0$, (b) $\Omega' = 0.25$, (c) $\Omega' = 0.5$, (d) $\Omega' = 0.75$, (e) $\Omega' = 1.0$, and (f) $\Omega' = 1.25$. Here, $f_{\text{bar}} = 0.68$ Hz, $f_m = 0$ Hz, $\phi_m = 0$, and the other parameters used are the same as Fig. 4(a).

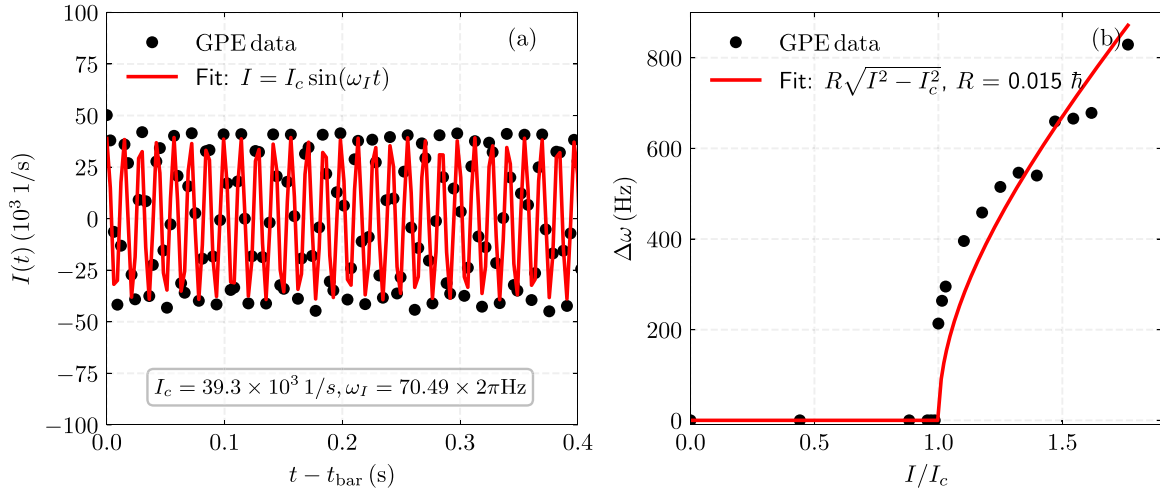


FIG. 9. Fitting procedure to obtain R , C , and I_c . (a) The tunneling current $I = dN/dt$ obtained from the GPE simulation is fitted by a sine function $I = I_c \sin(\omega_I t)$, which yields $I_c = 39.3 \times 10^3$ 1/s. (b) Magnitude of split in the peaks of the cavity output spectra $\Delta\omega$ is fitted by $\hbar\Delta\omega = \Delta\mu = R\sqrt{I^2 - I_c^2}$, which yields $R = 0.015 \hbar = 2 \times 10^{-3} \hbar$. We obtain C by calculating $n(t)/\Delta\mu(t)$ for $f_{\text{bar}} = 0.75$ Hz (greater than f_c) [28] and its mean value gives $C = 3.75 \text{ s}/\hbar = 23.55 \text{ s}/\hbar$. Here, $f_m = 0 \text{ Hz}$, $\phi_m = 0$, and the other parameters used are the same as Fig. 2.

for a certain barrier velocity, above f_c for different rotation rates Ω' [see Fig. 6(b)], which is similar to tracking the dc voltage across the SQUID as a function of external magnetic field [41]. The corresponding cavity output spectra are shown in Fig. 7, which shows a clear modulation in magnitude of peak splitting with the AQUID rotation rate with periodicity of one flux quantum of the ring Ω_0 .

APPENDIX B: RESISTIVELY AND CAPACITIVELY SHUNTED JUNCTION MODEL

The RCSJ model is used to simulate the characteristics of a Josephson junction by considering the nonzero capacitance of the Josephson junction arising from the repulsive atomic interactions and its resistance due to quasiparticle excitation when the junction current exceeds the critical current [27,28,53]. In this framework, using Kirchoff's circuit law, the total current flowing through a single junction can be written as

$$I = I_J + I_R + I_C, \\ I = I_c \sin(\Phi) - \frac{1}{R} \Delta\mu - C \frac{\Delta\mu}{dt}. \quad (\text{B1})$$

Here, $I_J = I_c \sin(\Phi)$ is the current through the junction, $I_R = \frac{1}{R} \Delta\mu$ is the current through the resistor with resistance R , and $I_C = C \frac{\Delta\mu}{dt}$ is the current through the capacitor with capacitance C . Φ and $\Delta\mu$ are the phase difference and the chemical potential difference at the junction. Using the expression $\Delta\mu = -\hbar\dot{\Phi}$, Eq. (B1) can be written as

$$\hbar C \ddot{\Phi} + \frac{\hbar}{R} \dot{\Phi} + I_c \sin(\Phi) = I \\ \Rightarrow \ddot{\Phi} + \frac{1}{RC} \dot{\Phi} + \frac{I_c}{\hbar C} \sin(\Phi) = \frac{I}{\hbar C}. \quad (\text{B2})$$

Using the nondimensionalization scheme $\tau = \omega_p t$, where $\omega_p = \sqrt{\frac{I_c}{\hbar C}}$, we arrive at

$$\frac{d^2\Phi}{d\tau^2} \omega_p^2 + \frac{1}{RC} \frac{d\Phi}{d\tau} \omega_p + \omega_p^2 \sin(\Phi) = \frac{I}{\hbar C} \\ \Rightarrow \frac{d^2\Phi}{d\tau^2} + \frac{1}{Q} \frac{d\Phi}{d\tau} + \sin(\Phi) = \frac{I}{I_c}, \quad (\text{B3})$$

where $Q = \omega_p RC$ is the quality factor, and the voltage across the junction is calculated via

$$V = \omega_p \left\langle \frac{d\Phi}{d\tau} \right\rangle. \quad (\text{B4})$$

For the AQUID device with two junctions (see Fig. 8 for the circuit diagram), considering the external rotation and flux quantization condition of the toroidal geometry, the circuit can be described by [31]

$$\frac{d^2\Phi_1}{d\tau^2} + \frac{1}{Q} \frac{d\Phi_1}{d\tau} + \sin(\Phi_1) = \frac{I_1}{I_c}, \\ \frac{d^2\Phi_2}{d\tau^2} + \frac{1}{Q} \frac{d\Phi_2}{d\tau} + \sin(\Phi_2) = \frac{I_2}{I_c}. \quad (\text{B5})$$

Defining $I_{1,2}/I_c = k_{1,2}$, $k_1 + k_2 = k$, $k_2 - k_1 = 2k_s$, and $1/Q = \beta_j$, and using the flux quantization condition

$$2\pi n = \Phi_1 - \Phi_2 - 2\pi \left(\Omega' + \frac{\beta}{2} k_s \right),$$

where $\beta = 2\pi I_c / N \Omega_0$, one can obtain the equations

$$\frac{d^2\Phi_1}{d\tau^2} + \beta_j \frac{d\Phi_1}{d\tau} + \sin(\Phi_1) = \frac{k}{2} - k_s, \\ \frac{d^2\Phi_2}{d\tau^2} + \beta_j \frac{d\Phi_2}{d\tau} + \sin(\Phi_2) = \frac{k}{2} + k_s, \quad (\text{B6})$$

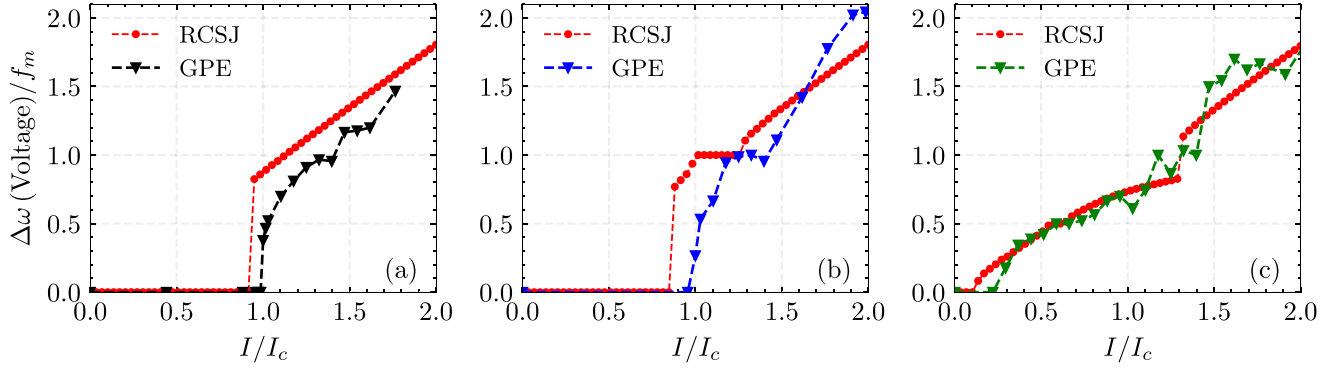


FIG. 10. Comparison between the results obtained from the GPE and RCSJ model for (a) the undriven case, (b) driven with $f_m = 90$ Hz, $\phi_m = 0.01$, $\Omega' = 0$ (that gives an integer Shapiro step), and (c) driven with $f_m = 90$ Hz, $\phi_m = 0.01$, $\Omega' = 0.5$ (that gives a half-integer Shapiro step). The other set of parameters used here is the same as that used in Fig. 2.

and

$$k_s = \frac{2}{\beta} \left[\frac{\Phi_1 - \Phi_2}{2\pi} - \Omega' - R \left(\frac{\Phi_1 - \Phi_2}{2\pi} - \Omega' \right) \right]. \quad (\text{B7})$$

Here, $R(\frac{\Phi_1 - \Phi_2}{2\pi} - \Omega')$ is the nearest integer function, and its subtraction yields the deviation from the quantized integer winding number, which drives the screening current. The system parameters R , C , and I_c are obtained via fitting the results obtained from GPE, and the fitting procedure is shown in Fig. 9.

To account for the periodic driving of the barrier ($\phi_{b1,2} = \pm(2\pi f_{\text{bar}}t \pm \phi_m \sin(2\pi f_m t))$) that generates the Shapiro steps, we have added the modulation current $\frac{I_m}{I_c} \sin(\frac{2\pi f_m \tau}{\omega_p})$ on the right-hand side of Eq. (B6), where f_m is the modulation frequency and $\frac{I_m}{I_c} = \frac{2\pi f_m \times \phi_m}{V_c}$. The voltage across the AQUID is calculated via

$$V = \frac{\omega_p}{2} \left\langle \frac{d\Phi_1}{d\tau} + \frac{d\Phi_2}{d\tau} \right\rangle. \quad (\text{B8})$$

We calculate voltage across the dc SQUID using Eq. (B8) for different bias currents and compare it to the numerically obtained results using GPE for the case of AQUID in Fig. 10. The slight mismatch between the RCSJ and GPE results is due

to the presence of damping and fluctuations in the BEC-cavity system.

APPENDIX C: CHARACTERISTICS OF THE SHAPIRO STEPS

1. Effect of the barrier rotation

In this Appendix, we present the variation of the amount of peak splitting ($\Delta\omega$) in the cavity output spectra as a function of the barrier velocity f_{bar} for different barrier rotation frequencies Ω' in Fig. 11. For $\Omega' = 0$, the case of zero rotation, we can see the integer Shapiro steps, which were seen earlier in the rectangular BEC cloud [28]. When we start rotating the junctions, we can see the emergence of additional steps whose height is around 0.75 [Fig. 11(b)], and the first step height becomes exactly 0.5 for $\Omega' = 0.5$, which is the half-integer Shapiro step [see Fig. 11(c)]. By further increasing the rotation frequency, the half-integer step vanishes and the integer step emerges for $\Omega' = 1$ [Fig. 11(e)]. Here, we have considered the rotation rate of AQUID in the range of 0–1. However, due to the periodic nature of the system, one can get similar results for Ω' beyond 1. In that way, we get the integer Shapiro steps for integer flux quantum and the half-integer steps for half-integer flux quantum.

However, due to the use of sinusoidal current-phase relationship [$I = I_c \sin(\Phi)$], the RCSJ model does not exhibit the fractional steps other than the half-integer ones. This confirms

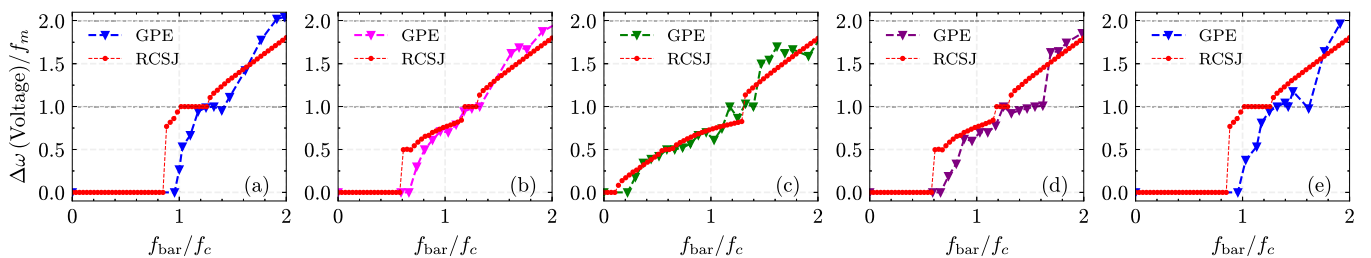


FIG. 11. Shapiro steps for $f_m = 90$ Hz, $\phi_m = 0.01$, and barrier rotation frequency (a) $\Omega' = 0.0$, (b) $\Omega' = 0.25$, (c) $\Omega' = 0.5$, (d) $\Omega' = 0.75$, and (e) $\Omega' = 1.0$. The other set of parameters used here is the same as used in Fig. 2.

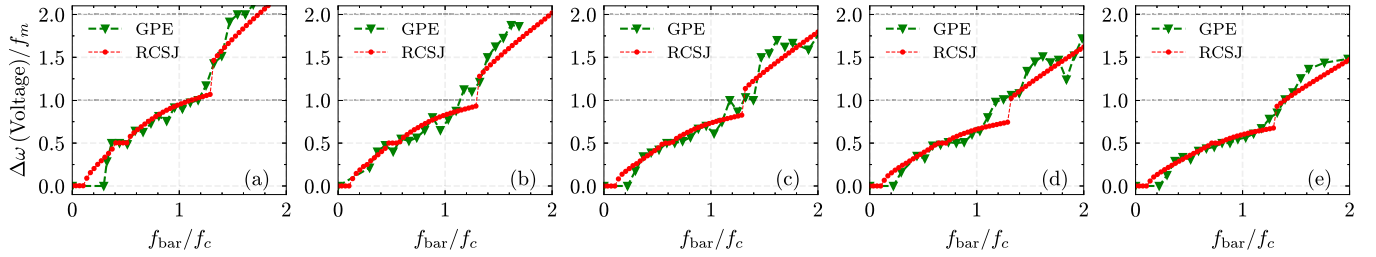


FIG. 12. Half-integer Shapiro steps for $\Omega' = 0.5$, $\phi_m = 0.01$, and barrier modulation frequency (a) $f_m = 70$ Hz, (b) $f_m = 80$ Hz, (c) $f_m = 90$ Hz, (d) $f_m = 100$ Hz, and (e) $f_m = 110$ Hz. The other set of parameters used here is the same as used in Fig. 2.

that the emergence of half-integer steps is solely a result of the rotation of AQUID.

2. Effect of the barrier modulation frequency

In this Appendix, we demonstrate the effect of increasing barrier modulation frequency on the half-integer Shapiro step. For $\Omega' = 0.5$, the variation of the amount of peak splitting ($\Delta\omega$) in the cavity output spectra as a function of the barrier velocity f_{bar} for different modulation frequencies f_m is shown in Fig. 12. As discussed in Fig. 11, we see the half-integer Shapiro steps for half-integer flux quantum $\Omega' = 0.5$, and we demonstrate it for a range of modulation frequency f_m with a constant barrier modulation period $t_{\text{bar}} = 0.092$ s. For all the cases, we can see the half-integer Shapiro steps (see Fig. 12) and this confirms that the emergence of half-integer Shapiro steps is solely due to the presence of half-integer rotational flux quantum ($\Omega' = 0.5$).

3. Cavity output spectra corresponding to integer and half-integer Shapiro steps

In this Appendix, we present the power spectrum of the phase quadrature of the cavity output field for different barrier velocities f_{bar} , keeping the barrier rotation rates and barrier modulation frequency constant in Fig. 13. For (a) $f_{\text{bar}} = 0.4$ Hz, (b) $f_{\text{bar}} = 0.45$ Hz, and (c) $f_{\text{bar}} = 0.5$ Hz, we can see that the magnitude of splitting of the peaks is the same, and they are nearly half of the barrier modulation frequency $f_m = 90$ Hz. These values contribute to the half-integer Shapiro steps in Fig. 12(c). For (d) $f_{\text{bar}} = 0.8$ Hz, (e) $f_{\text{bar}} = 0.9$ Hz, and (f) $f_{\text{bar}} = 0.95$ Hz, the magnitude of splitting of the peaks is nearly same as the barrier modulation frequency $f_m = 90$ Hz and contributes to the integer Shapiro steps in Fig. 12(c).

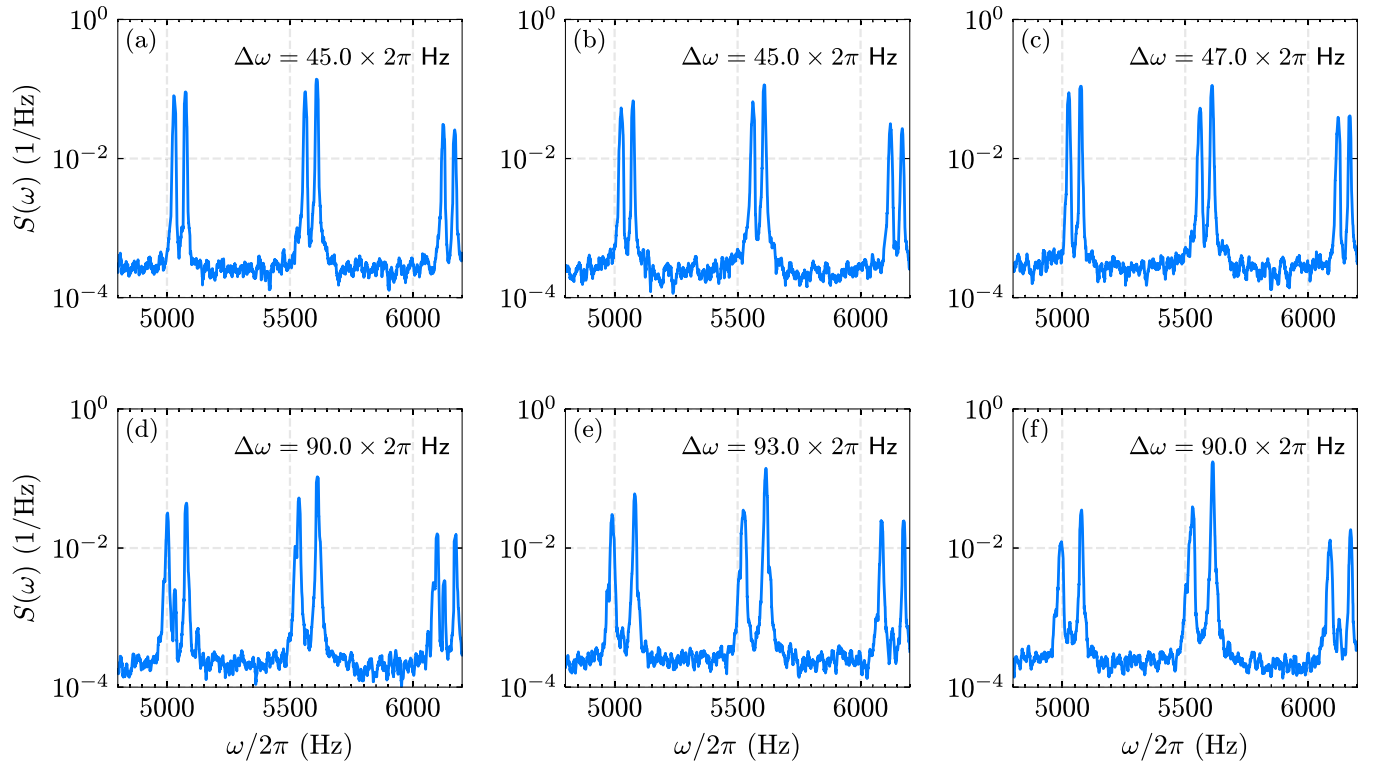


FIG. 13. Power spectrum of the phase quadrature of the cavity output field for barrier rotation rate $\Omega' = 0.5$, barrier modulation frequency $f_m = 90$ Hz, modulation amplitude $\phi_m = 0.01$, and barrier velocity (a) $f_{\text{bar}} = 0.4$ Hz, (b) $f_{\text{bar}} = 0.45$ Hz, (c) $f_{\text{bar}} = 0.5$ Hz, (d) $f_{\text{bar}} = 0.8$ Hz, (e) $f_{\text{bar}} = 0.9$ Hz, and (f) $f_{\text{bar}} = 0.95$ Hz. The upper row corresponds to the half-integer Shapiro steps, and the lower row corresponds to the integer Shapiro steps. The other set of parameters used here is the same as used in Fig. 2.

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